

Script-Controlled ACL: Restrict Record Access Based on Field Value

ServiceNow Micro Project Documentation

1. Introduction

- **Project Title:** Script-Controlled ACL: Restrict Record Access Based on Field Value
- **Platform:** ServiceNow (SmartBridge SkillWallet, MySkillWallet)
- **Estimated Duration:** 3 hours

Team Members:

Detail	Value	Role
Team ID	SWTID-2026-6385	-
Team Size	4	-
Team Leader	Yuvan Kumar S	Team Leader
Team Member	Yadesh Kannan	Team Member
Team Member	U Sivapriyan	Team Member
Team Member	Hasan Ahmed	Team Member

2. Project Description

This project demonstrates how to implement a script-controlled Access Control List (ACL) in ServiceNow to restrict users from viewing records unless they meet specific conditions. In this scenario, only users who hold the required custom role are allowed to view records where the Branch field is set to EEE, while administrators retain full access to every record.

A custom table, Institution Details (u_institution_details), stores student records across three branches: ECE, EEE and CSE. Four custom roles (bb1, bb2, bb3 and bb4) are created and layered on top of one another, so that each ACL operation (Read, Create, Write and Delete) is unlocked by its own role. The Read ACL uses an advanced script together with a data condition (Branch is EEE) to enforce record-level security, and the result is verified by impersonating the EEE User, a user without the roles, and the administrator.

Scenarios

Scenario 1: A user who holds the bb1 role opens the Institution Details list. Because the Read ACL checks the Branch field, only the EEE branch record is displayed, while ECE and CSE records stay hidden.

Scenario 2: A user who has no custom roles, such as Abel Tuter, opens the same list. The ACL denies access and the list shows "No records to display", protecting sensitive student data from unauthorized viewing.

Scenario 3: A system administrator opens the list. The script returns true for the admin role, so all records are visible regardless of branch. When a user holds bb1 to bb4, they can additionally create, edit and delete the EEE branch records.

3. Technical Architecture

Figure 1: Technical architecture of the Script-Controlled ACL implementation

Description: The solution is built entirely on the ServiceNow platform. Users and roles are managed under User Administration, and student data is stored in the custom table `u_institution_details`. Security is enforced by four record-level ACLs created under System Security → Access Control (ACL). When a user opens the table, ServiceNow evaluates the matching ACL for the requested operation (read, create, write or delete), checks the required role, the data condition and, for the Read ACL, the advanced script. Access is then either granted or denied. Impersonation is used to test the behavior as different users.

ER Diagram

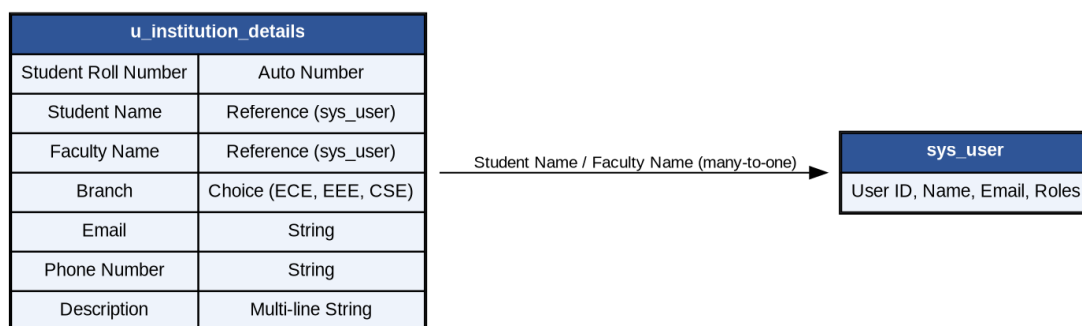


Figure 2: ER diagram of the Institution Details table

Description: The project uses one custom table, `u_institution_details`. Its Student Name and Faculty Name fields are reference fields that point to the existing `sys_user` table (many-to-one relationship). Branch is a choice field with the values ECE, EEE and CSE, and the Read ACL data condition is evaluated against this field. Student Roll Number is generated automatically (STU100, STU200, STU300 and so on).

4. Pre-requisites

- **ServiceNow Instance:** A personal developer instance (PDI) with admin access.
- **Basic ServiceNow Administration:** Users, roles, tables and fields.
- **Access Control (ACL) Concepts:** Operations, roles, data conditions and advanced scripts.
- **JavaScript Basics:** Used for the ACL script (`gs.hasRole`).
- **security_admin Role:** Required to create and modify ACLs; elevate the role before creating ACLs.
- **Impersonation:** Ability to impersonate users to verify the result.

5. Project Workflow

Milestone 1: Creation of Users and Roles

- Activity 1.1: Create the EEE User.
- Activity 1.2: Create the roles bb1, bb2, bb3 and bb4.
- Activity 1.3: Assign all roles to the EEE User.

Milestone 2: Table Creation

- Activity 2.1: Create the Institution Details table.

- Activity 2.2: Add the required fields.
- Activity 2.3: Create sample records for ECE, EEE and CSE.

Milestone 3: Creation of Access Control Lists (Read ACL)

- Activity 3.1: Elevate the security_admin role.
- Activity 3.2: Create the script-controlled Read ACL.

Milestone 4: Creation of Access Control Lists (Create ACL, bb2)

Milestone 5: Creation of Access Control Lists (Write ACL, bb3)

Milestone 6: Creation of Access Control Lists (Delete ACL, bb4)

Milestone 7: Impersonation and Verification

- Activity 7.1: Impersonate the EEE User.
- Activity 7.2: Impersonate a user without the role.
- Activity 7.3: Verify as Admin.
- Activity 7.4 to 7.6: Verify the Create, Write and Delete ACLs.

Milestone 8: Conclusion

Milestone 1: Creation of Users and Roles

In this milestone, a test user and four custom roles are created. The roles are later attached to the ACLs so that each operation (read, create, write and delete) can be controlled independently.

Activity 1.1: User Creation

- Log in to the ServiceNow instance with admin access.
- Navigate to User Administration → Users and click New.
- **User ID:** EEE User
- **First name:** EEE
- **Last name:** User
- **Email:** eeeuser@gmail.com
- Save the record.

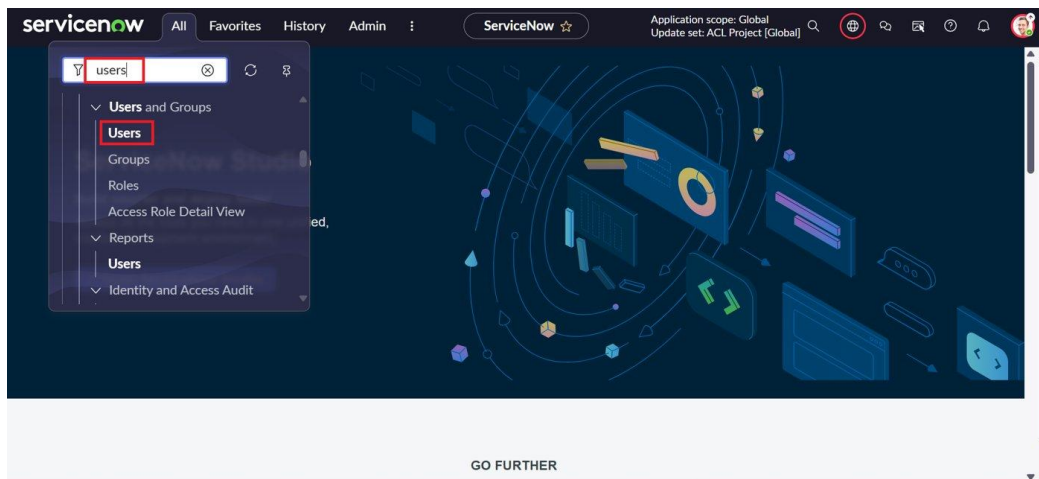


Figure 3: Navigating to User Administration → Users

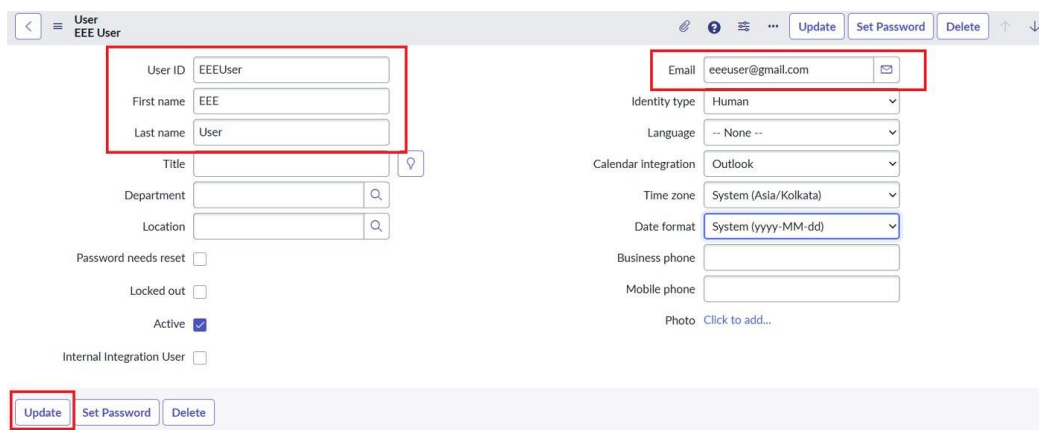


Figure 4: Creating the EEE User record

Activity 1.2: Role Creation

- Navigate to User Administration → Roles and click New.
- **Name:** bb1 (add a short description of the role purpose).
- Click Submit.
- Repeat the same procedure to create the roles bb2, bb3 and bb4.

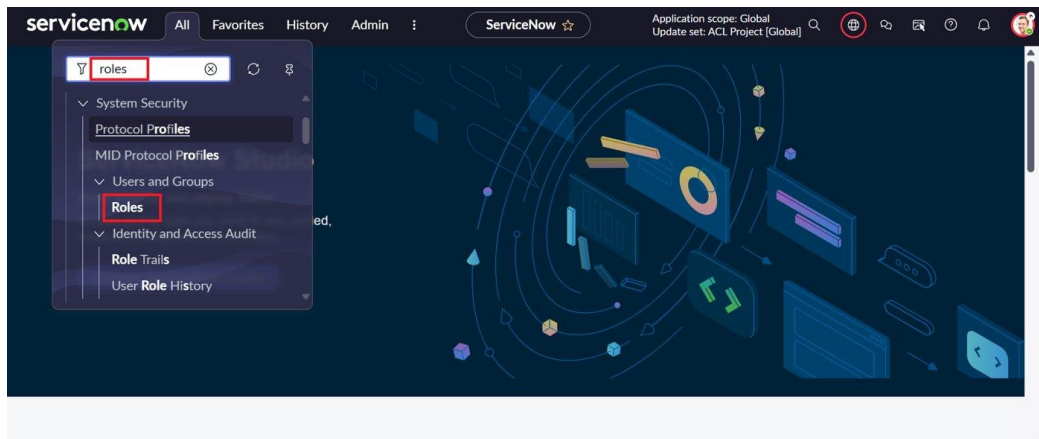


Figure 5: Navigating to User Administration → Roles

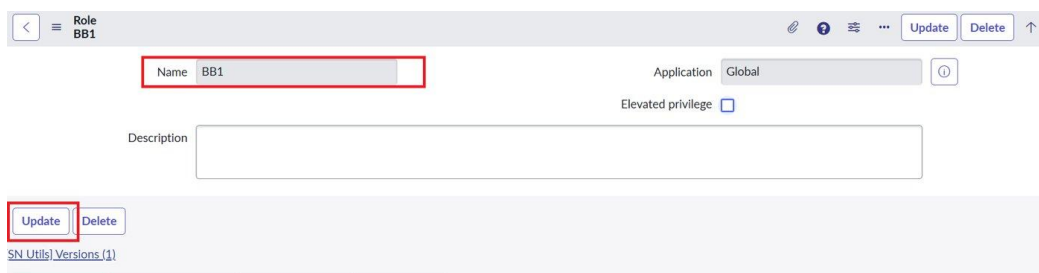


Figure 6: Creating the role bb1 (repeat for bb2, bb3 and bb4)

Activity 1.3: Assign Roles to the EEE User

- Open the EEE User record and scroll to the Roles related list.
- Click Edit and add bb1, bb2, bb3 and bb4 to the user.
- Click Save. The EEE User now holds bb1, bb2, bb3 and bb4, which are the roles later required by the Read, Create, Write and Delete ACLs.

Milestone 2: Table Creation

This milestone creates the custom table that holds the student records which the ACLs will protect.

Activity 2.1: Create the Table

- In the Application Navigator, search for System Definition → Tables and click New.
- **Label:** Institution Details
- **Name:** u_institution_details
- **Extends table:** false (none)
- Click Submit.

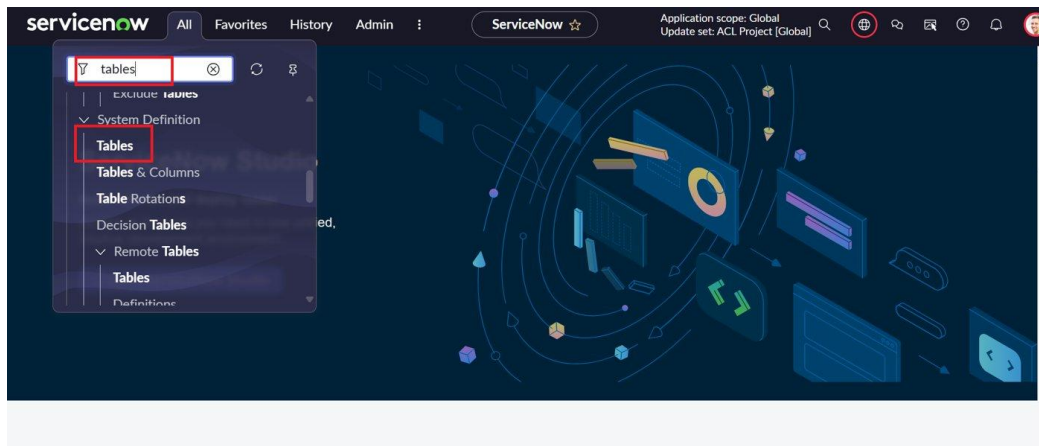


Figure 7: Navigating to System Definition → Tables

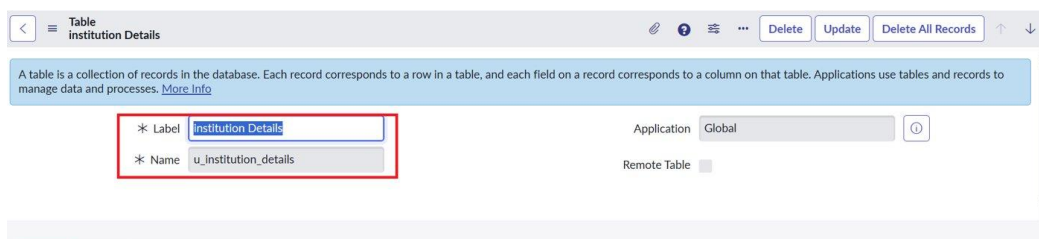


Figure 8: Creating the Institution Details table (u_institution_details)

Activity 2.2: Add Fields to the Table

Field	Type
Student Roll Number	Auto Number
Student Name	Reference (User)
Faculty Name	Reference (User)
Branch	Choice (ECE, EEE, CSE)
Email	String
Phone Number	String
Description	Multi-line String

Table Columns for text Search 1 to 13 of 13

Dictionary Entries

Column label	Type	Reference	Max length	Default value
Updates	Integer	(empty)	40	
Created by	String	(empty)	40	
Student Roll Number	String	(empty)	40	javascript:global.getNextObjNumberPadded();
Description	String	(empty)	1,000	
Updated by	String	(empty)	40	
Phone Number	String	(empty)	40	
Sys ID	Sys ID (GUID)	(empty)	32	
Created	Date/Time	(empty)	40	
Faculty name	Reference	User	32	
Branch	String	(empty)	40	
Updated	Date/Time	(empty)	40	
Student Name	Reference	User	32	
Email	String	(empty)	40	

Insert a new row...

Figure 9: Fields added to the u_institution_details table

Activity 2.3: Create Sample Records

Multiple records were created with different Branch values (ECE, EEE and CSE). When viewed as the administrator, the Institution Details list shows all three records.

servicenow All Favorites History Workspaces Institution Details Search Actions on selected rows... New

Branch	Description	Email	Faculty Name	Phone Number	Student Name	Student Roll Number
EEE			Allyson Gillispie		Alejandra Prenatt	STU200
CSE			Andrew Och		Amos Linnan	STU300
ECE			Abraham Lincoln		Abel Tuter	STU100

1 to 3 of 3

Figure 10: Institution Details list as admin showing the ECE, EEE and CSE records

Roll Number	Branch	Student Name	Faculty Name
STU100	ECE	Abel Tuter	Abraham Lincoln
STU200	EEE	Alejandra Prenatt	Allyson Gillispie
STU300	CSE	Amos Linnan	Andrew Och

Milestone 3: Creation of Access Control Lists (Read ACL)

This milestone creates the Read ACL, which restricts who can see records in u_institution_details. The ACL combines a required role, a data condition (Branch is EEE) and an advanced script.

Activity 3.1: Elevate the security_admin Role

Creating ACLs needs the security_admin role. Click the profile icon at the top right and select Elevate role, then enable security_admin.

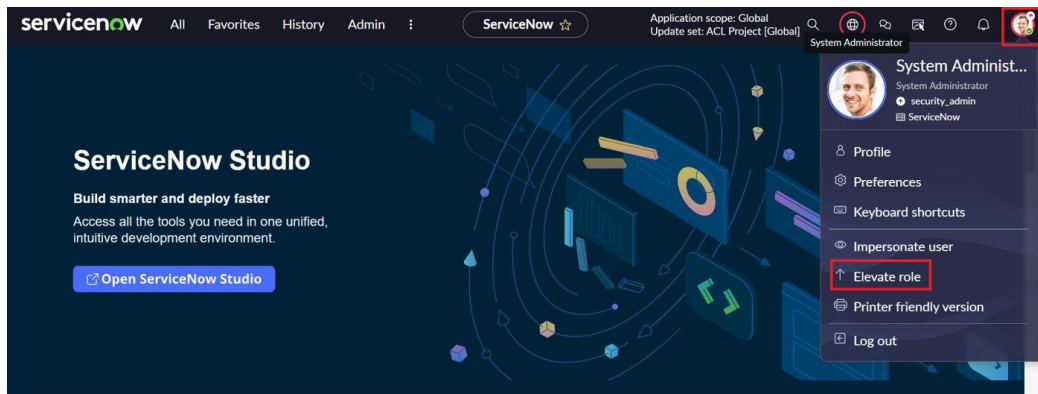


Figure 11: Profile menu showing the Elevate role option

Elevate role

Close dialog

Elevate a role by adding privileges, which end when you log out. [Learn more](#)

AVAILABLE ROLES

☒ security_admin

Grant modification access to High Security Settings, allow user to modify the Access Control List

Cancel

Update

Figure 12: Enabling the security_admin role and clicking Update

Activity 3.2: Create the Read ACL

- In the Application Navigator, search for System Security → Access Control (ACL) and click New.

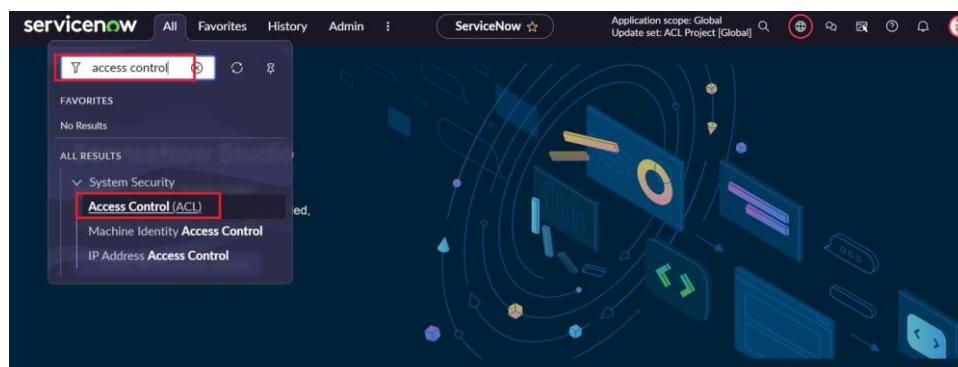


Figure 13: Opening System Security → Access Control (ACL)

servicenow							
Access Controls							
Name	Decision Type	Operation	Type	Active	Updated by	Updated	
\$allappsmgmt	Allow If	read	ui_page	true	admin	2019-02-20 01:02:07	
\$atf_page_inspector	Allow If	read	ui_page	true	admin	2020-03-20 15:16:15	
\$conversation-builder	Allow If	read	ui_page	true	admin	2019-06-14 15:13:38	
\$mycompanyappsmgmt	Allow If	read	ui_page	true	admin	2019-02-21 02:31:44	
\$products	Allow If	read	ui_page	true	admin	2020-03-10 03:19:22	
\$pwd_reset_serviceDesk	Allow If	read	ui_page	true	admin	2023-05-29 05:26:27	
\$sa_service_model_json	Allow If	read	ui_page	true	admin	2015-10-28 02:03:33	
\$sla_timeline	Allow If	read	ui_page	true	admin	2016-11-14 01:14:39	
\$sn-vn-web-client-app	Allow If	read	ui_page	true	admin	2018-03-06 10:20:09	
\$sn_global_search_results	Allow If	read	ui_page	true	admin	2020-06-15 10:43:57	
\$spd	Allow If	read	ui_page	true	admin	2016-01-19 12:53:23	
\$spm_availability_calculation_setting	Allow If	read	ui_page	true	admin	2024-02-18 06:04:39	
\$studio	Allow If	read	ui_page	true	admin	2016-01-09 16:21:38	
\$upgrade_client	Allow If	read	ui_page	true	admin	2025-11-10 16:44:04	
\$upgrade_monitor	Allow If	read	ui_page	true	admin	2025-09-29 22:40:08	

Figure 14: Access Controls list with the New button

- **Type:** record
- **Operation:** read
- **Name:** u_institution_details
- **Active:** true
- **Advanced:** true
- In the **Requires role** related list, add the custom role **bb1**.
- Add the data condition **Branch is EEE**.
- In the Script field, add the following script and click Submit:

```
(function () {
    // Allow admin users full access
    if (gs.hasRole('admin')) {
        return true;
    }

    // Allow only EEE branch users to see EEE records
    if (gs.hasRole('bb1')) {
        return true;
    }

    // Deny access for all others
    return false;
})();
```

Access Control
u_institution_details

* Type: record

* Operation: read

Decision Type: Allow If

Admin overrides: ☒

Protection policy: -- None --

* Name: institution Details [u_institution_details]

Description:

Applies To: No. of records matching the condition: 6

Add Filter Condition Add OR Clause

-- choose field -- -- oper -- -- value --

Application: Global

Active: ☒

Advanced: ☒

Figure 15: Read ACL form: type record, operation read, Advanced checked

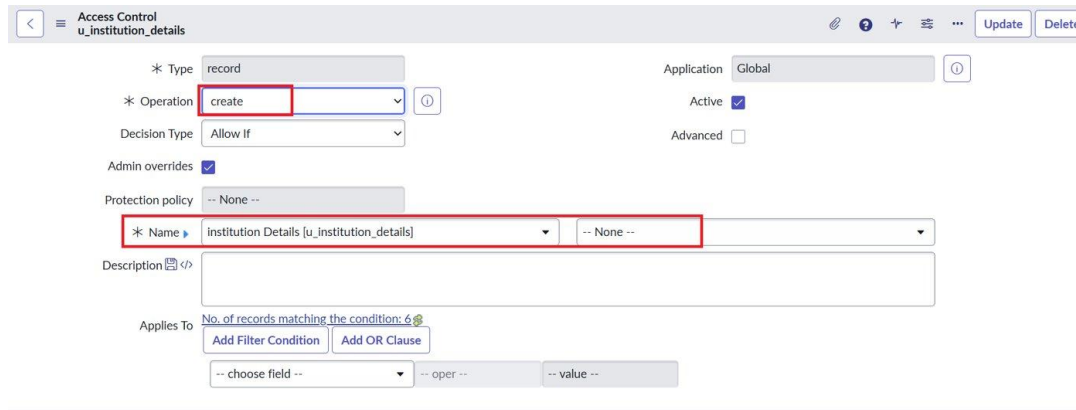


Figure 16: Data condition (Branch is EEE) and the advanced script

How the script works: The admin role is checked first and always returns true, so administrators keep full access. A user with the bb1 role is allowed through, and the data condition (Branch is EEE) then limits which rows that user can see. Every other user falls through to return false and is denied.

The Read ACL is tested by impersonating users in Milestone 7, after all four ACLs have been created.

Milestone 4: Creation of Access Control Lists (Create ACL)

The Create ACL controls who sees the New button and can insert new records in the table.

Activity 4.1: Create the ACL

- From Access Controls, click New.
- **Type:** record
- **Operation:** create
- **Name:** Institution Details [u_institution_details]
- **Active:** true

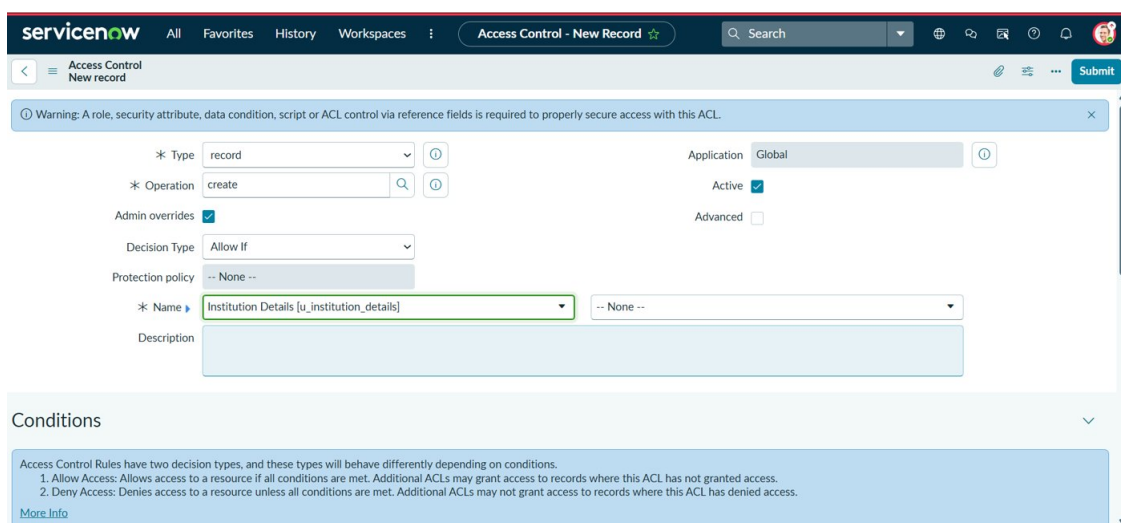


Figure 17: Create ACL form for u_institution_details

- In the **Requires role** related list, add the role **bb2**. No data condition is needed.

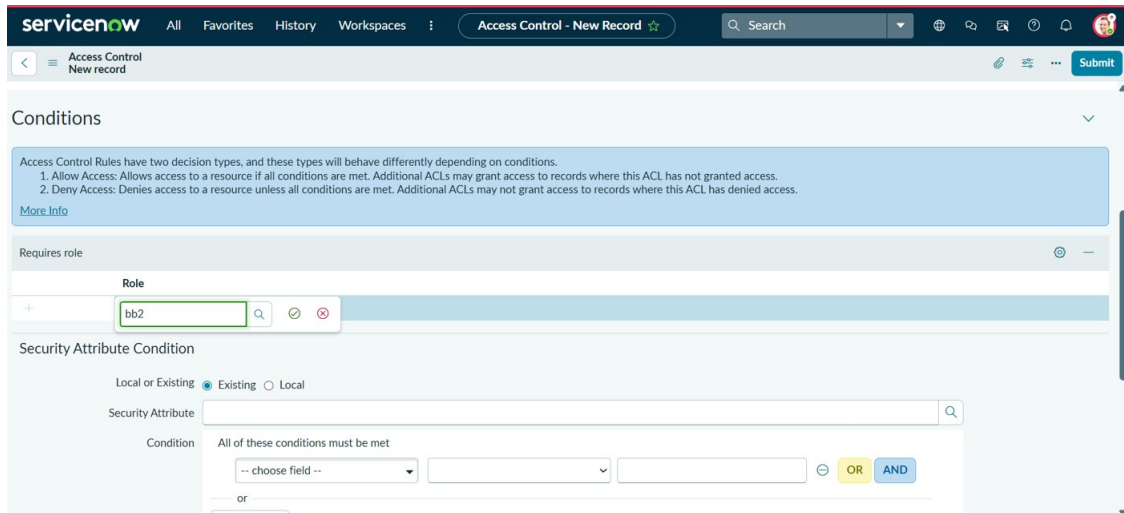


Figure 18: Adding the bb2 role to the Create ACL

- Save the ACL record. The Verify Security Rules dialog confirms the Create (Allow) rule on u_institution_details.

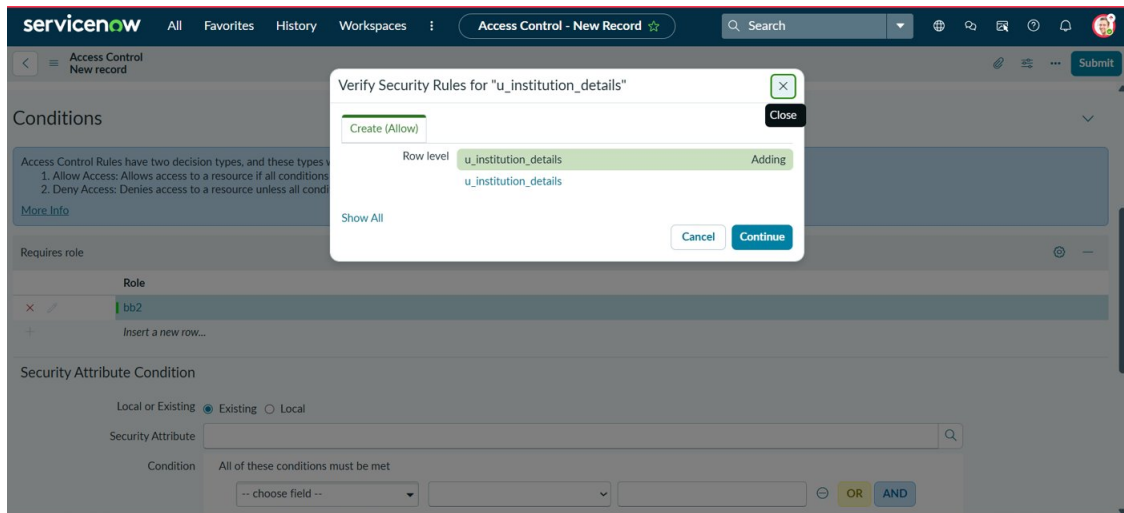


Figure 19: Verify Security Rules dialog showing Create (Allow) with role bb2

Expected Result

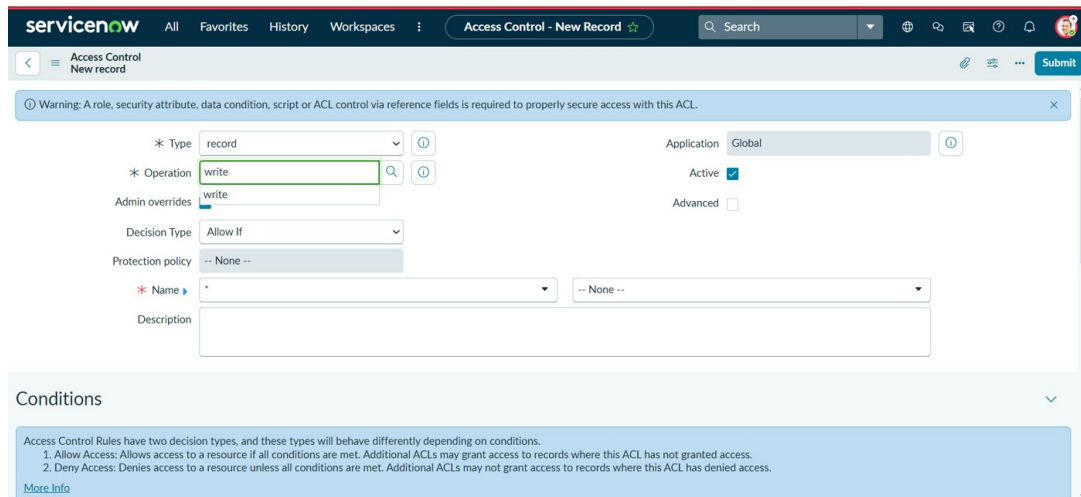
A user with the bb1 and bb2 roles can view only the EEE branch records and now also sees the New button on the list. This is verified by impersonation in Milestone 7.

Milestone 5: Creation of Access Control Lists (Write ACL)

The Write ACL controls who can edit records in the table.

Activity 5.1: Create the ACL

- From Access Controls, click New.
- **Type:** record
- **Operation:** write
- **Name:** u_institution_details
- **Active:** true



Warning: A role, security attribute, data condition, script or ACL control via reference fields is required to properly secure access with this ACL.

* Type: record

* Operation: write

Admin overrides: write

Decision Type: Allow If

Protection policy: -- None --

* Name: -- None --

Description:

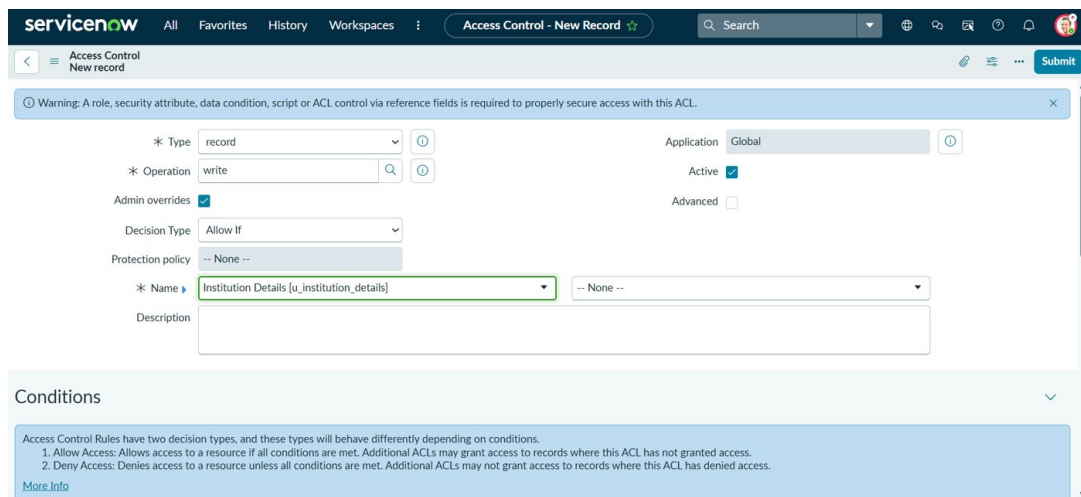
Conditions

Access Control Rules have two decision types, and these types will behave differently depending on conditions.

1. Allow Access: Allows access to a resource if all conditions are met. Additional ACLs may grant access to records where this ACL has not granted access.
2. Deny Access: Denies access to a resource unless all conditions are met. Additional ACLs may not grant access to records where this ACL has denied access.

[More Info](#)

Figure 20: Entering the write operation on a new ACL



Warning: A role, security attribute, data condition, script or ACL control via reference fields is required to properly secure access with this ACL.

* Type: record

* Operation: write

Admin overrides: ☒

Decision Type: Allow If

Protection policy: -- None --

* Name: Institution Details [u_institution_details]

Description:

Conditions

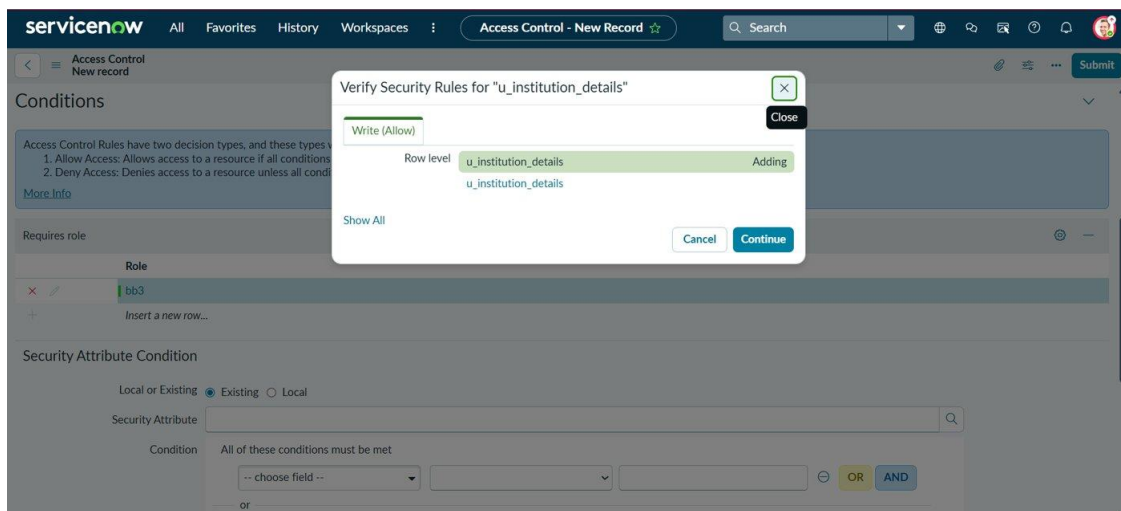
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1. Allow Access: Allows access to a resource if all conditions are met. Additional ACLs may grant access to records where this ACL has not granted access.
2. Deny Access: Denies access to a resource unless all conditions are met. Additional ACLs may not grant access to records where this ACL has denied access.

[More Info](#)

Figure 21: Write ACL form for Institution Details [u_institution_details]

- In the **Requires role** related list, add the role **bb3**. No data condition is needed.
- Save the ACL record.



Verify Security Rules for "u_institution_details"

Write (Allow)

Row level: u_institution_details Adding

u_institution_details

Show All

Cancel Continue

Conditions

Access Control Rules have two decision types, and these types will behave differently depending on conditions.

1. Allow Access: Allows access to a resource if all conditions are met. Additional ACLs may grant access to records where this ACL has not granted access.
2. Deny Access: Denies access to a resource unless all conditions are met. Additional ACLs may not grant access to records where this ACL has denied access.

[More Info](#)

Requires role

Role: bb3

Insert a new row...

Security Attribute Condition

Local or Existing: ☒ Existing ☐ Local

Security Attribute:

Condition: All of these conditions must be met

-- choose field --

OR AND

Figure 22: Verify Security Rules dialog showing Write (Allow) with role bb3

Expected Result

A user with the bb1, bb2 and bb3 roles can view and edit the EEE branch records, along with the New button. This is verified by impersonation in Milestone 7.

Milestone 6: Creation of Access Control Lists (Delete ACL)

The Delete ACL controls who can remove records from the table.

Activity 6.1: Create the ACL

- From Access Controls, click New.
- **Type:** record
- **Operation:** delete

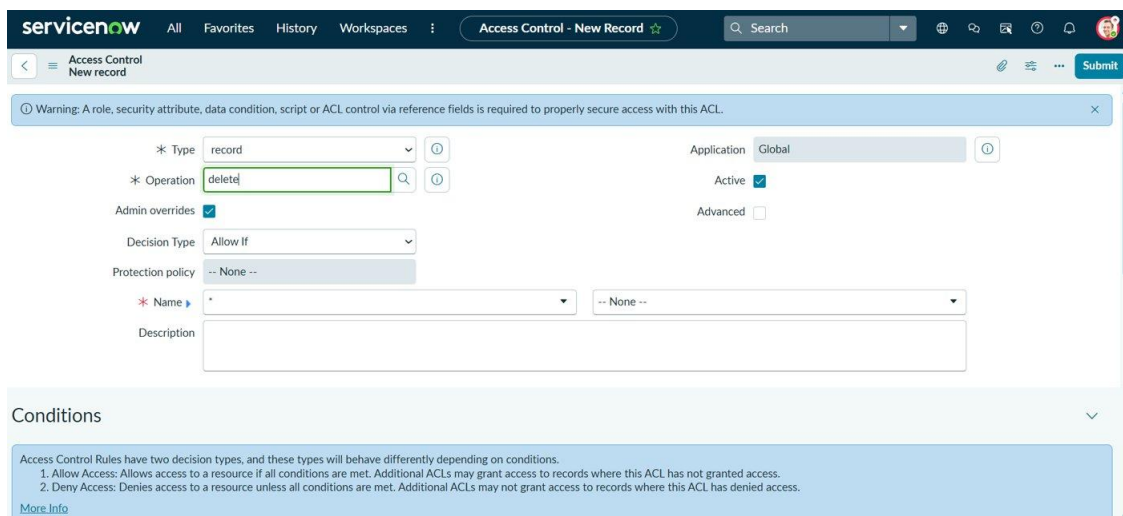


Figure 23: Entering the delete operation on a new ACL

- **Name:** Institution Details [u_institution_details]
- **Active:** true

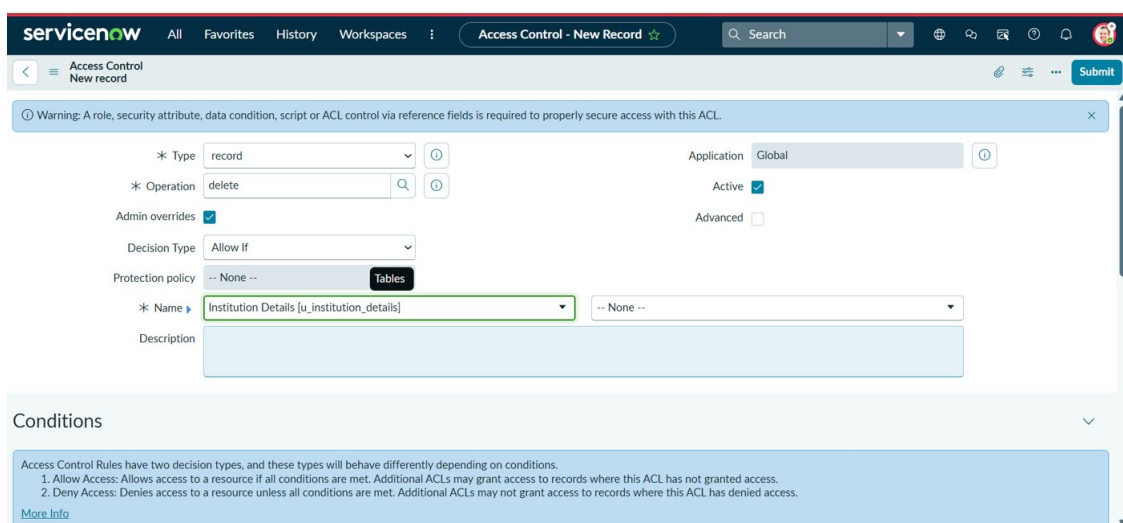


Figure 24: Delete ACL with the table name selected

- In the **Requires role** related list, add the role **bb4**. No data condition is needed.

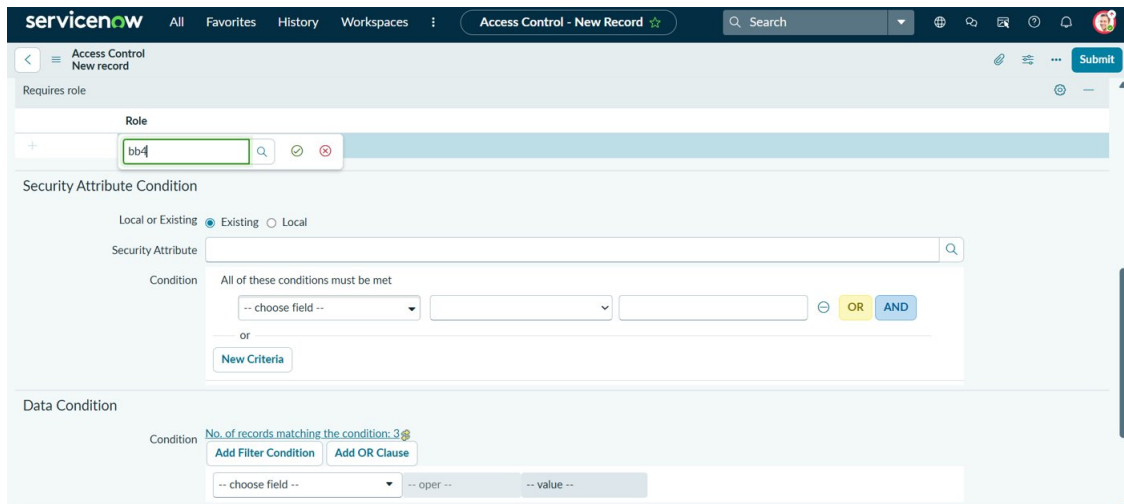


Figure 25: Adding the bb4 role to the Delete ACL

- Save the ACL record. The Verify Security Rules dialog confirms the Delete (Allow) rule on u_institution_details.

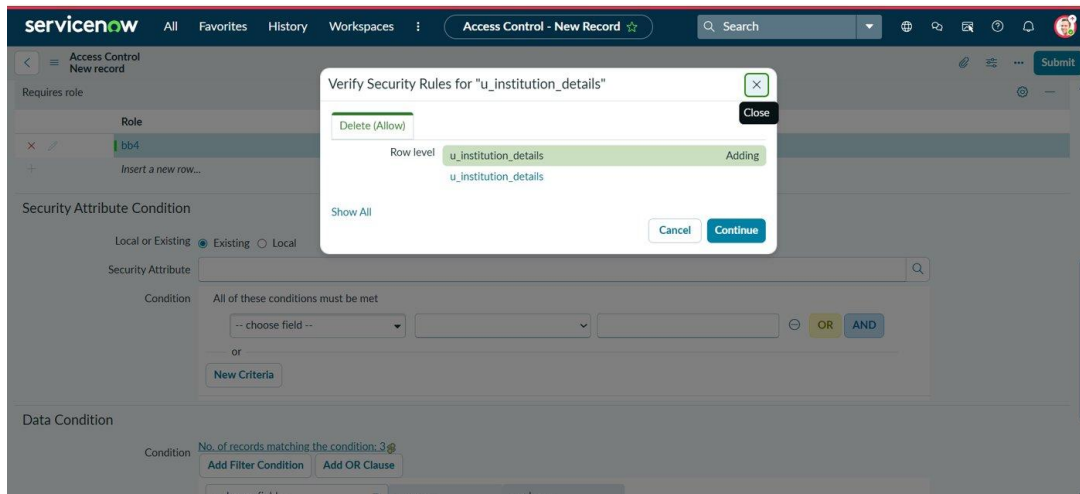
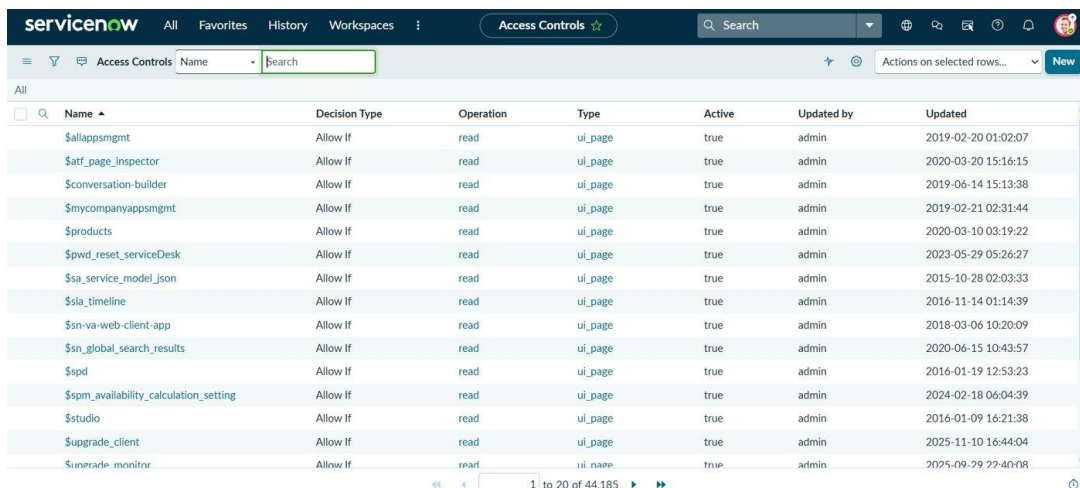


Figure 26: Verify Security Rules dialog showing Delete (Allow) with role bb4

Activity 6.2: Review the ACL List

Open System Security → Access Control (ACL) to review the Access Controls list and confirm the new ACLs for u_institution_details.



Name	Decision Type	Operation	Type	Active	Updated by	Updated
\$allappsmgmt	Allow If	read	ui_page	true	admin	2019-02-20 01:02:07
\$atf_page_inspector	Allow If	read	ui_page	true	admin	2020-03-20 15:16:15
\$conversation-builder	Allow If	read	ui_page	true	admin	2019-06-14 15:13:38
\$mycompanyappsmgmt	Allow If	read	ui_page	true	admin	2019-02-21 02:31:44
\$products	Allow If	read	ui_page	true	admin	2020-03-10 03:19:22
\$pwd_reset_serviceDesk	Allow If	read	ui_page	true	admin	2023-05-29 05:26:27
\$sa_service_model_json	Allow If	read	ui_page	true	admin	2015-10-28 02:03:33
\$sla_timeline	Allow If	read	ui_page	true	admin	2016-11-14 01:14:39
\$sn-va-web-client-app	Allow If	read	ui_page	true	admin	2018-03-06 10:20:09
\$sn_global_search_results	Allow If	read	ui_page	true	admin	2020-06-15 10:43:57
\$spd	Allow If	read	ui_page	true	admin	2016-01-19 12:53:23
\$spm_availability_calculation_setting	Allow If	read	ui_page	true	admin	2024-02-18 06:04:39
\$studio	Allow If	read	ui_page	true	admin	2016-01-09 16:21:38
\$upgrade_client	Allow If	read	ui_page	true	admin	2025-11-10 16:44:04
\$untrade_monitor	Allow If	read	ui_name	true	admin	2025-09-29 22:40:08

Figure 27: Access Controls list after creating the ACLs

Expected Result

Users with the bb1, bb2, bb3 and bb4 roles can see, create and edit, and also delete the EEE branch records. This is verified by impersonation in Milestone 7.

ACL Summary

Operation	Role	Condition / Script	Effect
Read	bb1	Advanced script + Branch is EEE	View only EEE records (admin sees all)
Create	bb2	None	New button available
Write	bb3	None	Edit records
Delete	bb4	None	Delete records

Milestone 7: Impersonation and Verification

After all four ACLs were created, the behavior was verified by impersonating different users and opening the Institution Details list. To open the list, copy the table name `u_institution_details.list`, paste it in the All menu filter and press Enter.

Activity 7.1: Impersonate the EEE User (Read ACL)

Click the profile icon at the top right and choose Impersonate user.

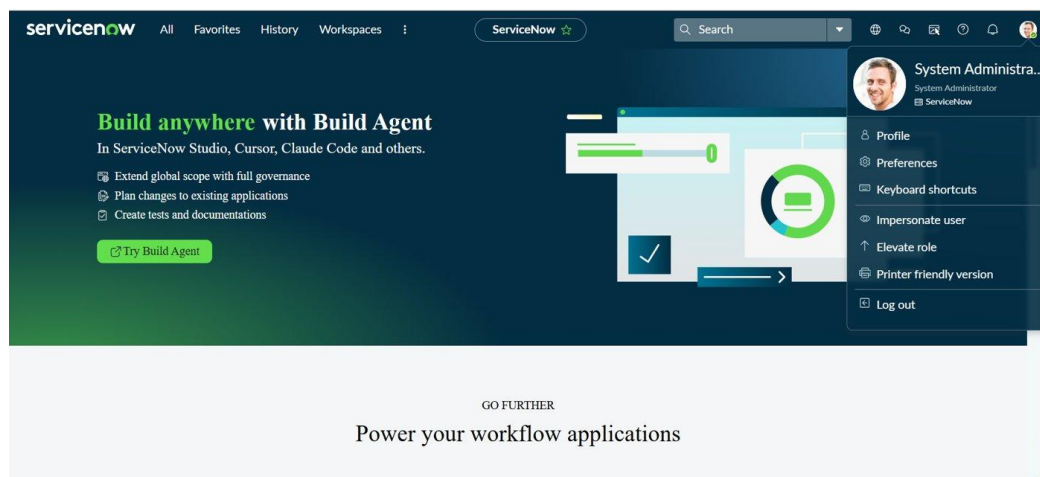


Figure 28: Profile menu showing Impersonate user and Elevate role options

Select EEE User from the recent impersonations and click Impersonate user.

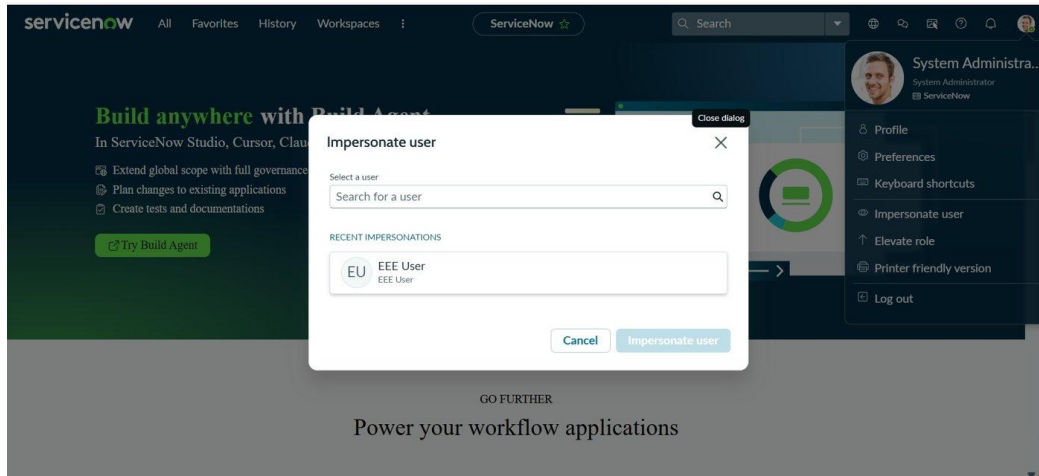


Figure 29: Impersonate user dialog with EEE User in the recent list

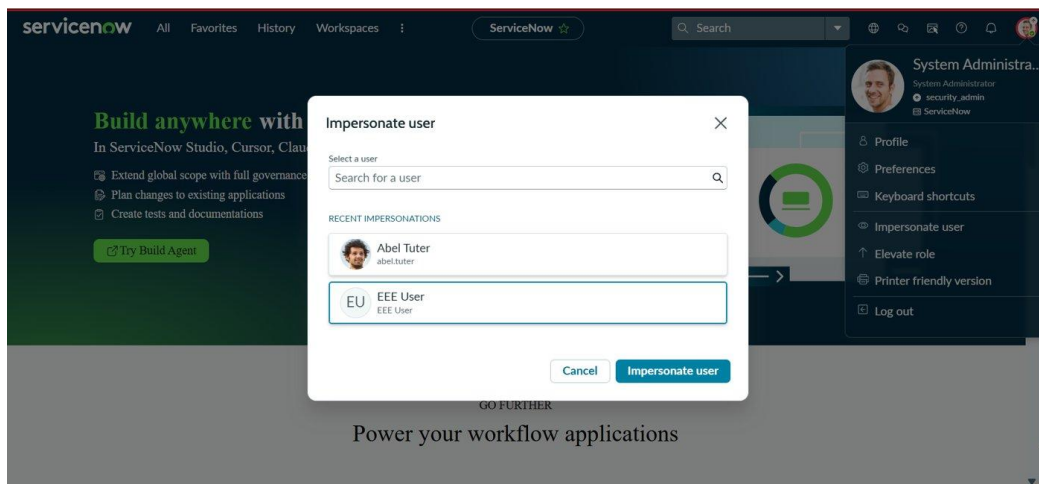


Figure 30: EEE User selected for impersonation (security_admin elevated)

The EEE User sees only the EEE branch record (STU200). The ECE and CSE records are hidden.

servicenow						
Institution Details						
Branch						
Branch	Description	Email	Faculty Name	Phone Number	Student Name	Student Roll Number
EEE			Allyson Gillispie		Alejandra Prenatt	STU200

Figure 31: EEE User sees only the EEE branch record

Activity 7.2: Impersonate a User Without the Role

Impersonate Abel Tuter, who has none of the custom roles, and open the same list.

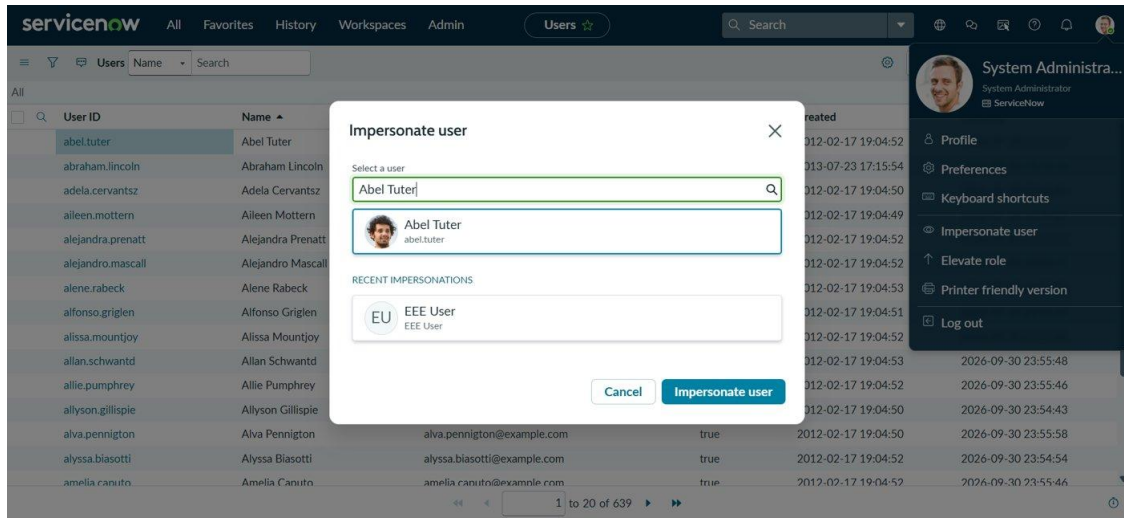


Figure 32: Impersonating Abel Tuter from the Users list

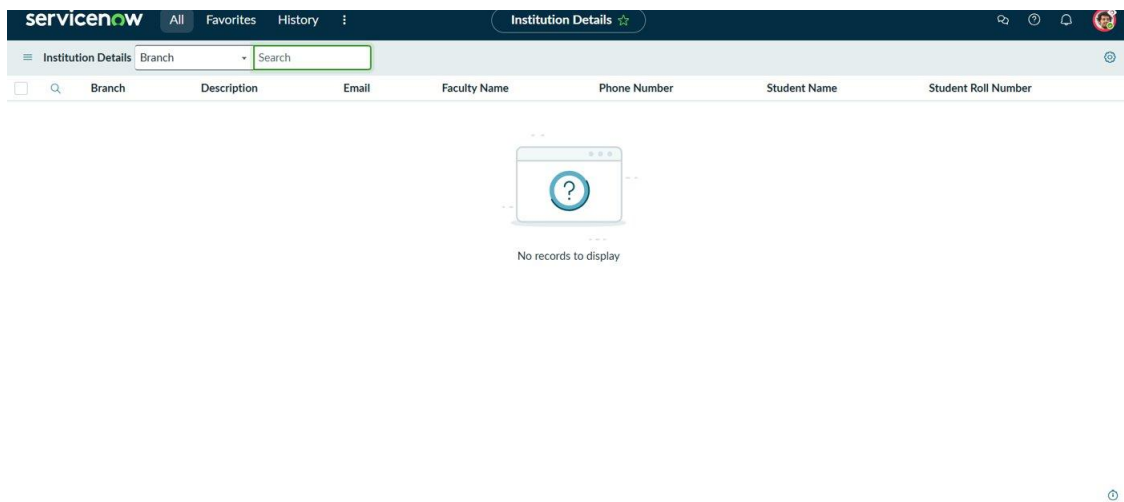


Figure 33: User without the role sees "No records to display"

Activity 7.3: Verify as Admin

Return to the administrator account and open the list again. All three records (ECE, EEE and CSE) are visible regardless of branch.

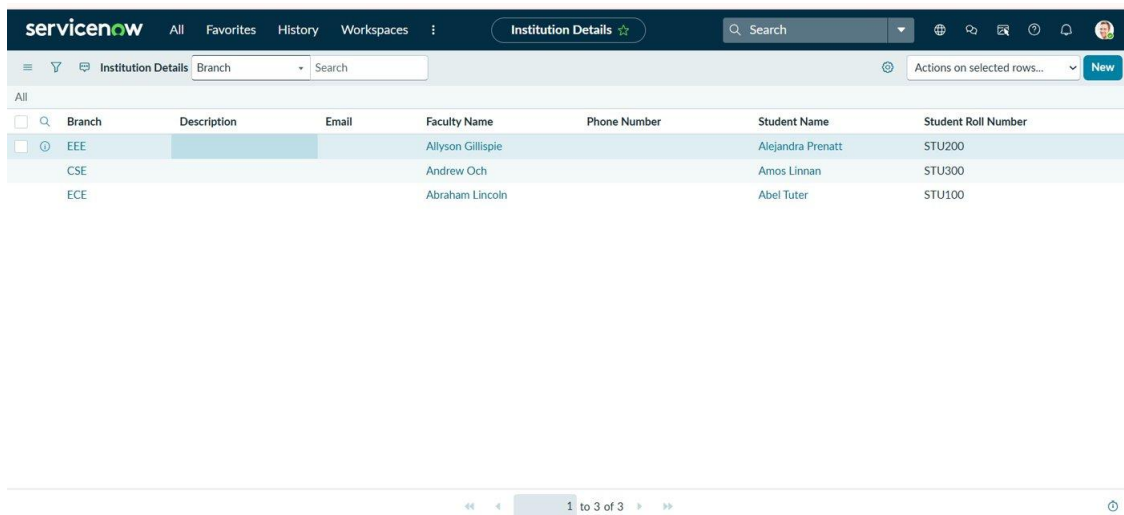
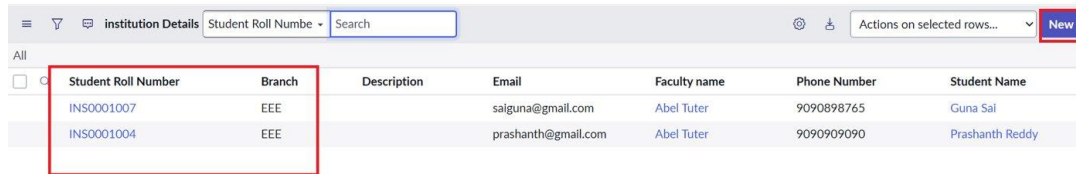


Figure 34: Admin user sees all records

Activity 7.4: Verify the Create ACL (bb1 + bb2)

As a user with the bb1 and bb2 roles, the list shows only the EEE branch records and the New button is available.

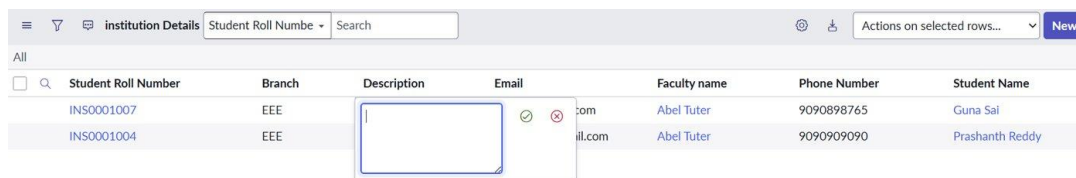


Student Roll Number	Branch	Description	Email	Faculty name	Phone Number	Student Name
INS0001007	EEE		saiguna@gmail.com	Abel Tuter	9090898765	Guna Sai
INS0001004	EEE		prashanth@gmail.com	Abel Tuter	9090909090	Prashanth Reddy

Figure 35: EEE records with the New button visible for bb1 + bb2

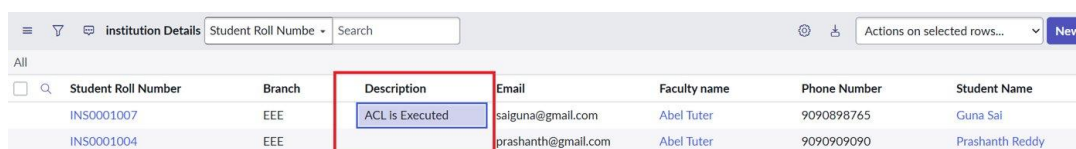
Activity 7.5: Verify the Write ACL (bb1 + bb2 + bb3)

With the bb3 role added, the EEE records can also be edited. For example, the Description field becomes editable and can be saved.



Student Roll Number	Branch	Description	Email	Faculty name	Phone Number	Student Name
INS0001007	EEE		saiguna@gmail.com	Abel Tuter	9090898765	Guna Sai
INS0001004	EEE		prashanth@gmail.com	Abel Tuter	9090909090	Prashanth Reddy

Figure 36: Editing the Description field of an EEE record



Student Roll Number	Branch	Description	Email	Faculty name	Phone Number	Student Name
INS0001007	EEE	ACL Is Executed	saiguna@gmail.com	Abel Tuter	9090898765	Guna Sai
INS0001004	EEE		prashanth@gmail.com	Abel Tuter	9090909090	Prashanth Reddy

Figure 37: Edited value saved on the EEE records

Activity 7.6: Verify the Delete ACL (bb1 + bb2 + bb3 + bb4)

With the bb4 role added, an EEE record can be deleted. Select a record, open Actions on selected rows, choose Delete and confirm.

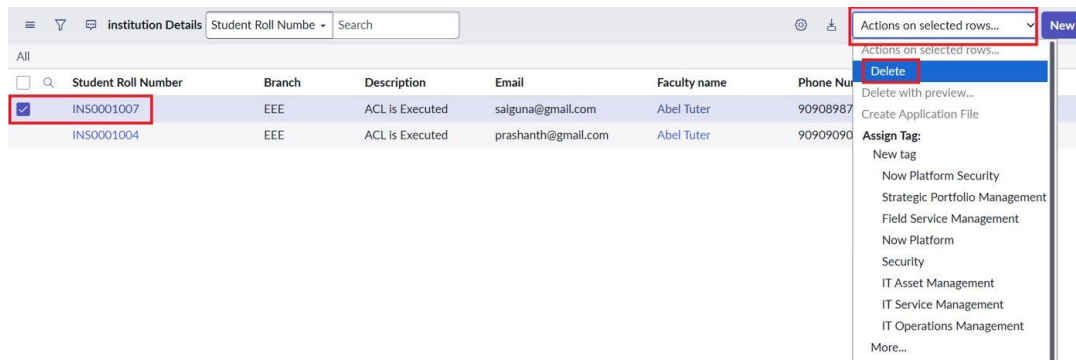


Figure 38: Selecting Delete from Actions on selected rows

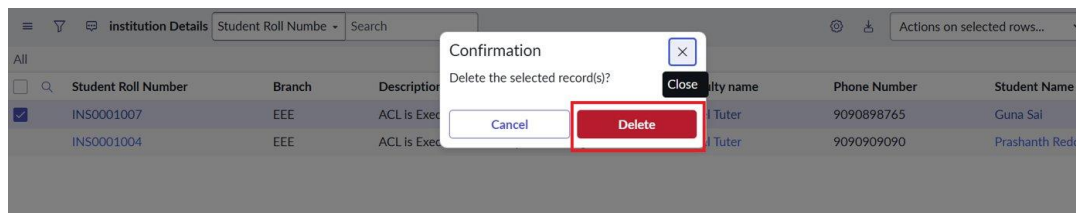


Figure 39: Delete confirmation dialog

Verification Summary

User / Roles	Result
User with bb1 role	Can view only EEE branch records
User without the role	Cannot view any records
Admin user	Can view all records regardless of branch
bb1 + bb2	Can view EEE records and use the New button
bb1 + bb2 + bb3	Can view and edit EEE records, along with New
bb1 + bb2 + bb3 + bb4	Can see, create, edit and delete EEE records

Milestone 8: Conclusion

This project showed how script-controlled READ, WRITE, CREATE and DELETE ACLs enforce record-level security in ServiceNow. By combining roles (bb1 to bb4), a data condition on the Branch field and an advanced script, access to the Institution Details table was restricted so that EEE users see only EEE branch records, users without the roles see nothing, and administrators keep full access. Impersonation confirmed each behavior. Together, these ACLs protect sensitive data from unauthorized viewing, creation, modification and deletion across forms and lists, and the same pattern can be reused for other tables and field-based restrictions.